

mielke-geometrodynamics: residuals

equations measured 2026-09-13T09:05:20Z against the model of 2026-09-12

Corrected (24)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00120 (was)	60	■0.651	$d\,\Sigma_{\alpha}\simeq(e_{\alpha}\downharpoonleft F)\wedge j$	$d\Sigma_{\alpha}\simeq(e_{\alpha}\downharpoonleft F)\wedge j$	$d\Sigma_{\alpha}\simeq(e_{\alpha}\downharpoonleft F)\wedge j$
mielke-geometrodynamics_E00120 (now)	60	—	$d\,\Sigma_{\alpha}\simeq(e_{\alpha}\rfloor F)\wedge j$	$d\Sigma_{\alpha}\simeq(e_{\alpha}\rfloor F)\wedge j$	$d\Sigma_{\alpha}\simeq(e_{\alpha}\rfloor F)\wedge j$
basis: census / census					
mielke-geometrodynamics_E00292 (was)	100	■0.165	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \downharpoonleft \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \downharpoonleft \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft R^{\gamma}_{\beta} \right) \wedge I^{\beta}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \downharpoonleft \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \downharpoonleft \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \downharpoonleft \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft R^{\gamma}_{\beta} \right) \wedge I^{\beta}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \downharpoonleft \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_E00292 (now)	100	—	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} \end{aligned}$	
basis: census / census					
mielke-geometrodynamics_E00293 (was)	101	■0.052	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	
mielke-geometrodynamics_E00293 (now)	101	—	$\xi \rfloor L \equiv \left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	$\xi \rfloor L \equiv \left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	
basis: census / census					
mielke-geometrodynamics_E00294 (was)	101	■0.510	$\Sigma_{\alpha} = e_{\alpha} \downharpoonleft L - \left(e_{\alpha} \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} - \left(e_{\alpha} \lrcorner \Psi \right) \wedge \frac{\partial L}{\partial \Psi} .$	$\Sigma_{\alpha} = e_{\alpha} \downharpoonleft L - \left(e_{\alpha} \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} - \left(e_{\alpha} \lrcorner \Psi \right) \wedge \frac{\partial L}{\partial \Psi} .$	
Conf. MathPix confidence: ■≥0.9 ■0.5-0.9 ■<0.5 —no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_E08294 (now)	101	—	$\left.\Sigma_{\alpha}=\left\lfloor e_{\alpha}\right\rfloor L-\left\lfloor e_{\alpha}\right\rfloor D \Psi\right) \wedge \frac{\partial L}{\partial D \Psi}-\left(e_{\alpha} \lrcorner \Psi\right) \wedge \frac{\partial L}{\partial \Psi} .$	$\Sigma_{\alpha}=e_{\alpha} \rfloor L-\left(e_{\alpha} \rfloor D \Psi\right) \wedge \frac{\partial L}{\partial D \Psi}-\left(e_{\alpha} \lrcorner \Psi\right) \wedge \frac{\partial L}{\partial \Psi} .$	$\Sigma_{\alpha}=e_{\alpha} \rfloor L-\left(e_{\alpha} \rfloor D \Psi\right) \wedge \frac{\partial L}{\partial D \Psi}-\left(e_{\alpha} \lrcorner \Psi\right) \wedge \frac{\partial L}{\partial \Psi} .$
basis: census / census					
mielke-geometrodynamics_E08296 (was)	101	■0.847	$F_{\alpha}=\left\lfloor e_{\alpha}\right\rfloor D \Psi\right) \frac{\delta L}{\delta \Psi}+\left(-1\right)^p\left(e_{\alpha} \lrcorner \Psi\right) \wedge D \frac{\delta L}{\delta \Psi}$	$F_{\alpha}=\left(e_{\alpha} \rfloor D \Psi\right) \frac{\delta L}{\delta \Psi}+\left(-1\right)^p\left(e_{\alpha} \rfloor \Psi\right) \wedge D \frac{\delta L}{\delta \Psi}$	$F_{\alpha}=\left(e_{\alpha} \rfloor D \Psi\right) \frac{\delta L}{\delta \Psi}+\left(-1\right)^p\left(e_{\alpha} \rfloor \Psi\right) \wedge D \frac{\delta L}{\delta \Psi}$
mielke-geometrodynamics_E08296 (now)	101	—	$F_{\alpha}=\left\lfloor e_{\alpha}\right\rfloor D \Psi\right) \frac{\delta L}{\delta \Psi}+\left(-1\right)^p\left(e_{\alpha} \lrcorner \Psi\right) \wedge D \frac{\delta L}{\delta \Psi}$	$F_{\alpha}=\left(e_{\alpha} \rfloor D \Psi\right) \frac{\delta L}{\delta \Psi}+\left(-1\right)^p\left(e_{\alpha} \rfloor \Psi\right) \wedge D \frac{\delta L}{\delta \Psi}$	$F_{\alpha}=\left(e_{\alpha} \rfloor D \Psi\right) \frac{\delta L}{\delta \Psi}+\left(-1\right)^p\left(e_{\alpha} \rfloor \Psi\right) \wedge D \frac{\delta L}{\delta \Psi}$
basis: census / census					
mielke-geometrodynamics_E08385 (was)	103	■0.793	$\left.\varepsilon_{\mathrm{RC}}:=\xi^{\alpha} \Sigma_{\alpha}+\left(e_{\beta} \lrcorner \widehat{D} \xi^{\gamma}\right) \tau^{\beta}{}_{\gamma},\right\}$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha} \Sigma_{\alpha}+\left(e_{\beta} \rfloor \widehat{D} \xi^{\gamma}\right) \tau^{\beta}{}_{\gamma},$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha} \Sigma_{\alpha}+\left(e_{\beta} \rfloor \widehat{D} \xi^{\gamma}\right) \tau^{\beta}{}_{\gamma},$
mielke-geometrodynamics_E08385 (now)	103	—	$\left.\varepsilon_{\mathrm{RC}}:=\xi^{\alpha} \Sigma_{\alpha}+\left(e_{\beta} \lrcorner \widehat{D} \xi^{\gamma}\right) \tau^{\beta}{}_{\gamma},\right\}$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha} \Sigma_{\alpha}+\left(e_{\beta} \rfloor \widehat{D} \xi^{\gamma}\right) \tau^{\beta}{}_{\gamma},$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha} \Sigma_{\alpha}+\left(e_{\beta} \rfloor \widehat{D} \xi^{\gamma}\right) \tau^{\beta}{}_{\gamma},$
basis: census / census					
mielke-geometrodynamics_E08378 (was)	128	■0.136	$D D \vartheta^{\alpha}=R_{\beta}^{\alpha} \wedge \vartheta^{\beta}=0, \stackrel{\circ}{D} \Pi_{\alpha}=-\stackrel{\circ}{R}_{\alpha}^{\beta} \wedge \Pi_{\beta}=\pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Sigma_{\beta} \cong 0 .$	$D D \vartheta^{\alpha}=R_{\beta}^{\alpha} \wedge \vartheta^{\beta}=0, \stackrel{\circ}{D} \Pi_{\alpha}=-\stackrel{\circ}{R}_{\alpha}^{\beta} \wedge \Pi_{\beta}=\pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Sigma_{\beta} \cong 0 .$	$D D \vartheta^{\alpha}=R_{\beta}^{\alpha} \wedge \vartheta^{\beta}=0, \stackrel{\circ}{D} \Pi_{\alpha}=-\stackrel{\circ}{R}_{\alpha}^{\beta} \wedge \Pi_{\beta}=\pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Sigma_{\beta} \cong 0 .$
mielke-geometrodynamics_E08378 (now)	128	—	$D D \vartheta^{\alpha}=R_{\beta}^{\alpha} \wedge \vartheta^{\beta}=0, \stackrel{\circ}{D} \Pi_{\alpha}=-\stackrel{\circ}{R}_{\alpha}^{\beta} \wedge \Pi_{\beta}=\pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Sigma_{\beta} \cong 0 .$	$D D \vartheta^{\alpha}=R_{\beta}^{\alpha} \wedge \vartheta^{\beta}=0, \stackrel{\circ}{D} \Pi_{\alpha}=-\stackrel{\circ}{R}_{\alpha}^{\beta} \wedge \Pi_{\beta}=\pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Sigma_{\beta} \cong 0 .$	$D D \vartheta^{\alpha}=R_{\beta}^{\alpha} \wedge \vartheta^{\beta}=0, \stackrel{\circ}{D} \Pi_{\alpha}=-\stackrel{\circ}{R}_{\alpha}^{\beta} \wedge \Pi_{\beta}=\pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Sigma_{\beta} \cong 0 .$
basis: census / census					
mielke-geometrodynamics_E08381 (was)	129	■0.588	$\left.\stackrel{\circ}{D} \Pi_{\alpha}:=D \Pi_{\alpha} \pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Pi_{\beta},\right\}$	$\stackrel{\circ}{D} \Pi_{\alpha}:=D \Pi_{\alpha} \pm \frac{i}{2} \ell^2\left(e_{\alpha} \rfloor \Pi^{\beta}\right) \wedge \Pi_{\beta},$	$\stackrel{\circ}{D} \Pi_{\alpha}:=D \Pi_{\alpha} \pm \frac{i}{2} \ell^2\left(e_{\alpha} \rfloor \Pi^{\beta}\right) \wedge \Pi_{\beta},$
mielke-geometrodynamics_E08381 (now)	129	—	$\left.\stackrel{\circ}{D} \Pi_{\alpha}:=D \Pi_{\alpha} \pm \frac{i}{2} \ell^2\left(e_{\alpha} \lrcorner \Pi^{\beta}\right) \wedge \Pi_{\beta},\right\}$	$\stackrel{\circ}{D} \Pi_{\alpha}:=D \Pi_{\alpha} \pm \frac{i}{2} \ell^2\left(e_{\alpha} \rfloor \Pi^{\beta}\right) \wedge \Pi_{\beta},$	$\stackrel{\circ}{D} \Pi_{\alpha}:=D \Pi_{\alpha} \pm \frac{i}{2} \ell^2\left(e_{\alpha} \rfloor \Pi^{\beta}\right) \wedge \Pi_{\beta},$
basis: census / census					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ0382 (was)	129	■ 0.087	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+i\stackrel{\scriptscriptstyle{(\mp)}}{A}{}^{\beta}{}_{\beta},\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}{}^{\beta}{}_{\beta}:=\pm\frac{1}{2}\ell^2e_{\alpha}\downarrow\Pi^{\beta},$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+i\stackrel{\scriptscriptstyle{(\mp)}}{A}{}^{\beta},\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}{}^{\beta}{}_{\beta}:=\pm\frac{1}{2}\ell^2e_{\alpha}\downarrow\Pi^{\beta},$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+i\stackrel{\scriptscriptstyle{(\mp)}}{A}_{\alpha}{}^{\beta},\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}_{\alpha}{}^{\beta}:=\pm\frac{1}{2}\ell^2e_{\alpha}\downarrow\Pi^{\beta},$
mielke-geometrodynamics_EQ0382 (now)	129	—	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+i\stackrel{\scriptscriptstyle{(\mp)}}{A}{}^{\beta}{}_{\beta},\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}{}^{\beta}{}_{\beta}:=\pm\frac{1}{2}\ell^2e_{\alpha}\downarrow\Pi^{\beta},$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+i\stackrel{\scriptscriptstyle{(\mp)}}{A}{}^{\beta},\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}{}^{\beta}{}_{\beta}:=\pm\frac{1}{2}\ell^2e_{\alpha}\downarrow\Pi^{\beta},$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_{\alpha}{}^{\beta}:=\Gamma_{\alpha}{}^{\beta}+i\stackrel{\scriptscriptstyle{(\mp)}}{A}_{\alpha}{}^{\beta},\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}_{\alpha}{}^{\beta}:=\pm\frac{1}{2}\ell^2e_{\alpha}\downarrow\Pi^{\beta},$
basis: census / census					
mielke-geometrodynamics_EQ0391 (was)	131	■ 0.066	$\begin{aligned}\Sigma_{\alpha}&:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\left.\left(\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\right.\right.\\&\quad\left.\left.-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}\right).\end{aligned}$	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\left.\left(e_{\alpha}\downarrow L-\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\right.\right.\\&\quad\left.\left.-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}\right).$	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=e_{\alpha}\downarrow L-\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\\&\quad-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
mielke-geometrodynamics_EQ0391 (now)	131	—	$\begin{aligned}\Sigma_{\alpha}&:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\left.\left(\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\right.\right.\\&\quad\left.\left.-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}\right).\end{aligned}$	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=\left.\left(e_{\alpha}\downarrow L-\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\right.\right.\\&\quad\left.\left.-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}\right).$	$\Sigma_{\alpha}:=\frac{\delta L}{\delta\vartheta^{\alpha}}=e_{\alpha}\downarrow L-\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\\&\quad-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}.$
basis: census / census					
mielke-geometrodynamics_EQ0392 (was)	132	■ 0.552	$\left.\left(\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\right.\right.\\&\quad\left.\left.-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}\right).$	$n^{\alpha}\Sigma_{\alpha}=n\downarrow L-\left(n\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(n\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}.$	$n^{\alpha}\Sigma_{\alpha}=n\downarrow L-\left(n\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(n\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}.$
mielke-geometrodynamics_EQ0392 (now)	132	—	$\left.\left(\left(e_{\alpha}\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(e_{\alpha}\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}\right.\right.\\&\quad\left.\left.-\left(e_{\alpha}\downarrow T^{\beta}\right)\wedge\frac{\partial L}{\partial T^{\beta}}+D\frac{\partial L}{\partial T^{\alpha}}-\left(e_{\alpha}\downarrow R_{\beta}^{\gamma}\right)\wedge\frac{\partial L}{\partial R_{\beta}^{\gamma}}\right).$	$n^{\alpha}\Sigma_{\alpha}=n\downarrow L-\left(n\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(n\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}.$	$n^{\alpha}\Sigma_{\alpha}=n\downarrow L-\left(n\downarrow D\Psi\right)\wedge\frac{\partial L}{\partial D\Psi}-\left(n\downarrow\Psi\right)\wedge\frac{\partial L}{\partial\Psi}.$
basis: census / census					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_132 EQ0393 (was)		■0.382	$\begin{aligned} n^{\alpha} \Sigma_{\alpha} &= n \lrcorner L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \rfloor \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ &\cong n \rfloor L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}]. \end{aligned}$	$n^{\alpha} \Sigma_{\alpha} = n \lrcorner L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \rfloor \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ \cong n \rfloor L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$	$n^{\alpha} \Sigma_{\alpha} = n \rfloor L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \rfloor \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ \cong n \rfloor L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$
mielke-geometrodynamics_132 EQ0393 (now)		—	$\begin{aligned} n^{\alpha} \Sigma_{\alpha} &= n \rfloor L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \rfloor \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ &\cong n \rfloor L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}]. \end{aligned}$	$n^{\alpha} \Sigma_{\alpha} = n \rfloor L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \rfloor \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ \cong n \rfloor L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$	$n^{\alpha} \Sigma_{\alpha} = n \rfloor L - (\mathfrak{t}_n \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (n \rfloor \Psi) \wedge \frac{\delta L}{\delta \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}] \\ \cong n \rfloor L - \left(\ell_n \Psi + \Gamma_{\perp}^{\alpha \beta} \rho(I_{\alpha \beta}) \Psi \right) \wedge \frac{\partial L}{\partial D \Psi} + D[(n \rfloor \Psi) \wedge \frac{\partial L}{\partial D \Psi}].$
basis: census / census					
mielke-geometrodynamics_134 EQ0402 (was)		■0.042	$\begin{array}{ll} \text{q: } & \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta), \\ \textstyle{\pm} p: & \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = -i \ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}} \Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta), \\ \textstyle{\pm} p: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = -i \ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}} \Psi_{\parallel}(\vartheta),$	$q: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta), \\ \textstyle{\pm} p: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = -i \ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}} \Psi_{\parallel}(\vartheta),$
mielke-geometrodynamics_134 EQ0402 (now)		—	$\begin{array}{ll} \text{q: } & \stackrel{*}{\Psi}_{\parallel}(\vartheta) = \stackrel{*}{\Psi}_{\parallel}(\vartheta), \\ \textstyle{\pm} p: & \stackrel{(\pm)}{\Pi}_{\alpha} \Psi_{\parallel}(\vartheta) = -i \ell^2 \frac{\delta}{\delta \stackrel{*}{\vartheta}_{\alpha}} \Psi_{\parallel}(\vartheta), \end{array}$	$q: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta), \\ \textstyle{\pm} p: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = -i \ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}} \Psi_{\parallel}(\vartheta),$	$q: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta), \\ \textstyle{\pm} p: \quad \underline{*}^{\vartheta} \Psi_{\parallel}(\vartheta) = -i \ell^2 \frac{\delta}{\delta \underline{*}^{\vartheta}} \Psi_{\parallel}(\vartheta),$
basis: inferred / ink					
mielke-geometrodynamics_142 EQ0437 (was)		■0.532	$E_{\alpha} := \frac{\partial L_g}{\partial \vartheta^{\alpha}} = e_{\alpha} \lrcorner L_g + (e_{\alpha} \lrcorner T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \lrcorner R^{\beta \gamma}) \wedge H_{\beta \gamma},$	$E_{\alpha} := \frac{\partial L_g}{\partial \vartheta^{\alpha}} = e_{\alpha} \lrcorner L_g + (e_{\alpha} \lrcorner T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \lrcorner R^{\beta \gamma}) \wedge H_{\beta \gamma},$	$E_{\alpha} := \frac{\partial L_g}{\partial \vartheta^{\alpha}} = e_{\alpha} \lrcorner L_g + (e_{\alpha} \lrcorner T^{\beta}) \wedge H_{\beta} + (e_{\alpha} \lrcorner R^{\beta \gamma}) \wedge H_{\beta \gamma},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E08437 (now)	142	—	$E_{\alpha}:=\frac{\partial L_g}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L_g+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma},$	$E_{\alpha}:=\frac{\partial L_g}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L_g+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma},$	$E_{\alpha}:=\frac{\partial L_g}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L_g+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma},$
basis: census / census					
mielke-geometrodynamics_E08440 (was)	143	■ 0.288	$\left.D \Sigma_{\alpha}\right\lceil \left(e_{\alpha}\right\rfloor \downharpoonleft T^{\gamma}\right) \wedge \Sigma_{\gamma}+\left(e_{\alpha}\right\rfloor \downharpoonleft R^{\gamma \delta}\right) \wedge \tau_{\gamma \delta}$	$D \Sigma_{\alpha} \cong\left(e_{\alpha}\rfloor T^{\gamma}\right) \wedge \Sigma_{\gamma}+\left(e_{\alpha}\rfloor R^{\gamma \delta}\right) \wedge \tau_{\gamma \delta}$	$D \Sigma_{\alpha} \cong\left(e_{\alpha}\rfloor T^{\gamma}\right) \wedge \Sigma_{\gamma}+\left(e_{\alpha}\rfloor R^{\gamma \delta}\right) \wedge \tau_{\gamma \delta}$
mielke-geometrodynamics_E08440 (now)	143	—	$\left.D \Sigma_{\alpha}\right\lceil \left(e_{\alpha}\right\rfloor \downharpoonleft T^{\gamma}\right) \wedge \Sigma_{\gamma}+\left(e_{\alpha}\right\rfloor \downharpoonleft R^{\gamma \delta}\right) \wedge \tau_{\gamma \delta}$	$D \Sigma_{\alpha} \cong\left(e_{\alpha}\rfloor T^{\gamma}\right) \wedge \Sigma_{\gamma}+\left(e_{\alpha}\rfloor R^{\gamma \delta}\right) \wedge \tau_{\gamma \delta}$	$D \Sigma_{\alpha} \cong\left(e_{\alpha}\rfloor T^{\gamma}\right) \wedge \Sigma_{\gamma}+\left(e_{\alpha}\rfloor R^{\gamma \delta}\right) \wedge \tau_{\gamma \delta}$
basis: census / census					
mielke-geometrodynamics_E08472 (was)	155	■ 0.682	$\stackrel{+}{\Omega}^{\{\}}=0 \Longleftrightarrow \mathcal{R}_{i j}^{\{\}}=0 \Longleftrightarrow R_{i j}^{\{\}}=\Lambda^{\prime} g_{i j} .$	<i>(not rendered)</i>	$\stackrel{+}{\Omega}^{\{\}}=0 \Longleftrightarrow \mathcal{R}_{i j}^{\{\}}=0 \Longleftrightarrow R_{i j}^{\{\}}=\Lambda^{\prime} g_{i j} .$
mielke-geometrodynamics_E08472 (now)	155	—	$\stackrel{+}{\Omega}^{\{\}}=0 \Longleftrightarrow \mathcal{R}_{i j}^{\{\}}=0 \Longleftrightarrow R_{i j}^{\{\}}=\Lambda^{\prime} g_{i j} .$	$\stackrel{+}{\Omega}^{\{\}}=0 \Longleftrightarrow \mathcal{R}_{i j}^{\{\}}=0 \Longleftrightarrow R_{i j}^{\{\}}=\Lambda^{\prime} g_{i j} .$	$\stackrel{+}{\Omega}^{\{\}}=0 \Longleftrightarrow \mathcal{R}_{i j}^{\{\}}=0 \Longleftrightarrow R_{i j}^{\{\}}=\Lambda^{\prime} g_{i j} .$
basis: census / census					
mielke-geometrodynamics_E08570 (was)	184	■ 0.403	$E_{\alpha}:=\partial L / \partial \vartheta^{\alpha}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma}$	$E_{\alpha}:=\partial L / \partial \vartheta^{\alpha}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma}$	$E_{\alpha}:=\partial L / \partial \vartheta^{\alpha}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma}$
mielke-geometrodynamics_E08570 (now)	184	—	$E_{\alpha}:=\partial L / \partial \vartheta^{\alpha}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma}$	$E_{\alpha}:=\partial L / \partial \vartheta^{\alpha}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma}$	$E_{\alpha}:=\partial L / \partial \vartheta^{\alpha}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+\left(e_{\alpha}\rfloor R^{\beta \gamma}\right) \wedge H_{\beta \gamma}$
basis: census / census					
mielke-geometrodynamics_E08627 (was)	211	■ 0.271	$E_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+2(-1)^s\left(e_{\alpha}\rfloor R^{* \beta}\right) \wedge H_{\beta}^{\star},$	$E_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+2(-1)^s\left(e_{\alpha}\rfloor R^{* \beta}\right) \wedge H_{\beta}^{\star},$	$E_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+2(-1)^s\left(e_{\alpha}\rfloor R^{* \beta}\right) \wedge H_{\beta}^{\star},$
mielke-geometrodynamics_E08627 (now)	211	—	$E_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+2(-1)^s\left(e_{\alpha}\rfloor R^{* \beta}\right) \wedge H_{\beta}^{\star},$	$E_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+2(-1)^s\left(e_{\alpha}\rfloor R^{* \beta}\right) \wedge H_{\beta}^{\star},$	$E_{\alpha}:=\frac{\partial L}{\partial \vartheta^{\alpha}}=e_{\alpha}\rfloor L+\left(e_{\alpha}\rfloor T^{\beta}\right) \wedge H_{\beta}+2(-1)^s\left(e_{\alpha}\rfloor R^{* \beta}\right) \wedge H_{\beta}^{\star},$
basis: census / census					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_E00654 (was)	216	■ 0.274	$\left.\left.E_{\alpha}=e_{\alpha}\right\downarrow L+\left[\left(e_{\alpha}\right\downarrow T^{\beta}\right)+\left(-1\right)^s2\delta\ell\left(e_{\alpha}\right\downarrow R^{\star\beta}\right)\wedge H_{\beta}+\left(-1\right)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\right\downarrow R^{\star\beta}\right)\wedge\vartheta_{\beta},$	$E_{\alpha}=e_{\alpha}\downarrow L+\left[\left(e_{\alpha}\downarrow T^{\beta}\right)+\left(-1\right)^s2\delta\ell\left(e_{\alpha}\downarrow R^{\star\beta}\right)\right]\wedge H_{\beta}+\left(-1\right)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\downarrow R^{\star\beta}\right)\wedge\vartheta_{\beta},$	$E_{\alpha}=e_{\alpha}\downarrow L+\left[\left(e_{\alpha}\downarrow T^{\beta}\right)+\left(-1\right)^s2\delta\ell\left(e_{\alpha}\downarrow R^{\star\beta}\right)\right]\wedge H_{\beta}+\left(-1\right)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\downarrow R^{\star\beta}\right)\wedge\vartheta_{\beta},$
mielke-geometrodynamics_E00654 (now)	216	—	$\left.\left.E_{\alpha}=e_{\alpha}\right\rfloor L+\left[\left(e_{\alpha}\right\rfloor T^{\beta}\right)+\left(-1\right)^s2\delta\ell\left(e_{\alpha}\right\rfloor R^{\star\beta}\right)\wedge H_{\beta}+\left(-1\right)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\right\rfloor R^{\star\beta}\right)\wedge\vartheta_{\beta},$	$E_{\alpha}=e_{\alpha}\downarrow L+\left[\left(e_{\alpha}\downarrow T^{\beta}\right)+\left(-1\right)^s2\delta\ell\left(e_{\alpha}\downarrow R^{\star\beta}\right)\right]\wedge H_{\beta}+\left(-1\right)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\downarrow R^{\star\beta}\right)\wedge\vartheta_{\beta},$	$E_{\alpha}=e_{\alpha}\downarrow L+\left[\left(e_{\alpha}\downarrow T^{\beta}\right)+\left(-1\right)^s2\delta\ell\left(e_{\alpha}\downarrow R^{\star\beta}\right)\right]\wedge H_{\beta}+\left(-1\right)^s\frac{2\gamma}{\ell}\left(e_{\alpha}\downarrow R^{\star\beta}\right)\wedge\vartheta_{\beta},$
basis: census / census					
mielke-geometrodynamics_E00718 (was)	229	■ 0.066	$\begin{aligned} \Sigma_{\alpha} & \& \left.\left.: = e_{\alpha}\right\rfloor L_{\Psi}-\left(e_{\alpha}\right\rfloor \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial \Psi}-\left(e_{\alpha}\right\rfloor \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}} \\ & -\left(e_{\alpha}\right\rfloor D \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial D \Psi}-\left(e_{\alpha}\right\rfloor D \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial D \bar{\Psi}} ; \end{aligned}$	$\Sigma_{\alpha}:=e_{\alpha}\downarrow L_{\Psi}-\left(e_{\alpha}\downarrow \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial \Psi}-\left(e_{\alpha}\downarrow \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}}-\\ \left(e_{\alpha}\downarrow D \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial D \Psi}-\left(e_{\alpha}\downarrow D \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial D \bar{\Psi}};$	$\Sigma_{\alpha}:=e_{\alpha}\downarrow L_{\Psi}-\left(e_{\alpha}\downarrow \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial \Psi}-\left(e_{\alpha}\downarrow \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}}-\\ \left(e_{\alpha}\downarrow D \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial D \Psi}-\left(e_{\alpha}\downarrow D \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial D \bar{\Psi}};$
mielke-geometrodynamics_E00718 (now)	229	—	$\begin{aligned} \Sigma_{\alpha} & \& \left.\left.: = e_{\alpha}\right\rfloor L_{\Psi}-\left(e_{\alpha}\right\rfloor \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial \Psi}-\left(e_{\alpha}\right\rfloor \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}} \\ & -\left(e_{\alpha}\right\rfloor D \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial D \Psi}-\left(e_{\alpha}\right\rfloor D \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial D \bar{\Psi}} ; \end{aligned}$	$\Sigma_{\alpha}:=e_{\alpha}\downarrow L_{\Psi}-\left(e_{\alpha}\downarrow \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial \Psi}-\left(e_{\alpha}\downarrow \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}}-\\ \left(e_{\alpha}\downarrow D \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial D \Psi}-\left(e_{\alpha}\downarrow D \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial D \bar{\Psi}};$	$\Sigma_{\alpha}:=e_{\alpha}\downarrow L_{\Psi}-\left(e_{\alpha}\downarrow \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial \Psi}-\left(e_{\alpha}\downarrow \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial \bar{\Psi}}-\\ \left(e_{\alpha}\downarrow D \Psi\right) \wedge \frac{\partial L_{\Psi}}{\partial D \Psi}-\left(e_{\alpha}\downarrow D \bar{\Psi}\right) \wedge \frac{\partial L_{\Psi}}{\partial D \bar{\Psi}};$
basis: census / census					
mielke-geometrodynamics_E00842 (was)	272	■ 0.852	$\mu:=e_{\alpha}\left\downarrow \mu^{\alpha}=\frac{3}{4}^*j_5$	$\mu:=e_{\alpha}\downarrow \mu^{\alpha}=\frac{3}{4}^*j_5$	$\mu:=e_{\alpha}\downarrow \mu^{\alpha}=\frac{3}{4}^*j_5$
mielke-geometrodynamics_E00842 (now)	272	—	$\mu:=e_{\alpha}\left\rfloor \mu^{\alpha}=\frac{3}{4}^*j_5$	$\mu:=e_{\alpha}\downarrow \mu^{\alpha}=\frac{3}{4}^*j_5$	$\mu:=e_{\alpha}\downarrow \mu^{\alpha}=\frac{3}{4}^*j_5$
basis: census / census					
mielke-geometrodynamics_E00857 (was)	278	■ 0.925	$\left.\mathbf{\ell}_{\mathfrak{n}}\right\rfloor \mathscr{G}^A \cong n\right\rfloor \left(D \underline{\tau}^A\right),$	<i>(not rendered)</i>	$\mathbb{L}_n\mathscr{G}^A \cong n\downarrow \left(D \underline{\tau}^A\right),$
mielke-geometrodynamics_E00857 (now)	278	—	$\left.\mathbf{\ell}_{\mathfrak{n}}\right\rfloor \mathscr{G}^A \cong n\right\rfloor \left(D \underline{\tau}^A\right),$	$\mathbb{L}_n\mathscr{G}^A \cong n\downarrow \left(D \underline{\tau}^A\right),$	$\mathbb{L}_n\mathscr{G}^A \cong n\downarrow \left(D \underline{\tau}^A\right),$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
basis: census / census					
mielke-geometrodynamics_314 E08972 (was)	314	■ 0.153	$\stackrel{\textcircled{\text{S}}}{\alpha}:=\stackrel{\textcircled{\text{O}}}{\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega}$	$\stackrel{\textcircled{\text{S}}}{D}\alpha:=\stackrel{\textcircled{\text{O}}}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$	$\stackrel{\textcircled{\text{O}}}{D}\alpha:=\stackrel{\textcircled{\text{O}}}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$
mielke-geometrodynamics_314 E08972 (now)	314	—	$\overset{(1)}{\alpha}:=\overset{(0)}{\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega}$	$\overset{(1)}{D}\alpha:=\overset{(0)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$	$\overset{(0)}{D}\alpha:=\overset{(0)}{d}\alpha-\omega^g\wedge\alpha-\alpha\wedge\omega$
basis: inferred / ink					
mielke-geometrodynamics_100 F08431 (was)	100	—	$\mathrm{t}_{\xi}:=\xi\mathrel{\mathop{\mathrm{D}}\nolimits}\xi$	<i>(not rendered)</i>	covariant Lie derivative $\mathbb{L}_{\xi}:=\xi\lrcorner D+D\xi$ on M with respect to an arbitrary vector
mielke-geometrodynamics_100 F08431 (now)	100	—	$\mathrm{t}_{\xi}:=\xi\lfloor D+D\xi\rfloor$	$\mathbb{L}_{\xi}:=\xi\lrcorner D+D\xi$	covariant Lie derivative $\mathbb{L}_{\xi}:=\xi\lrcorner D+D\xi$ on M with respect to an arbitrary vector
basis: census / census					
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5-0.9 ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

Low confidence

Identifier	Page	Conf.	LaTeX source	Rendered	Image
mielke-geometrodynamics_47 E0064 (2.8.6) [conf 0.054]	47	■ 0.054 ● N -2	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$	$\begin{aligned} & \stackrel{*}{\pi}(\gamma_1) = -\frac{1}{2\pi i} \operatorname{Tr} \Omega \quad (=0) \\ & \stackrel{*}{\pi}(\gamma_2) = -\frac{1}{8\pi^2} \{ \operatorname{Tr}(\Omega \wedge \Omega) - \operatorname{Tr}(\Omega) \wedge \operatorname{Tr}(\Omega) \} \\ & \quad \left(= -\frac{1}{32\pi^2} \epsilon^{\alpha\beta\mu\nu} F_{\alpha\beta}^j F_{\mu\nu j} \sqrt{ g } d^4x \right) \\ & \stackrel{*}{\pi}(\gamma_k) = 0 \text{ for } k \geq 3. \end{aligned}$
mielke-geometrodynamics_133 E0480 (6.4.19) [conf 0.060]	133	■ 0.060 ● S +1	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{A}}_{\alpha} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{\mathcal{A}}}_{\alpha} = 0,$	$\stackrel{(\pm)}{\mathcal{G}}_{\alpha} := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = \stackrel{(\pm)}{D} \stackrel{(\pm)}{\underline{\mathcal{A}}}_{\alpha} = 0,$
mielke-geometrodynamics_168 E0529 (7.7.49) [conf 0.099]	168	■ 0.099 ●	$\begin{aligned} & L_{\text{Euler}} := (-1)^{\text{sig}+1} \operatorname{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ & = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ & \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \operatorname{Ric}_{\alpha\beta} \wedge {}^*\operatorname{Ric}^{\alpha\beta} + \frac{1}{2} \operatorname{Ric}_{\alpha}^{\alpha} \wedge {}^*\operatorname{Ric}_{\beta}^{\beta} \\ & \equiv -L_{\text{SKY}} \\ & \quad - 2 \left(\operatorname{Ric}_{\alpha\beta} \wedge {}^*\operatorname{Ric}^{\alpha\beta} - \frac{1}{4} \operatorname{Ric}_{\alpha}^{\alpha} \wedge {}^*\operatorname{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$\begin{aligned} & L_{\text{Euler}} := (-1)^{\text{sig}+1} \operatorname{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ & = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ & \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \operatorname{Ric}_{\alpha\beta} \wedge {}^*\operatorname{Ric}^{\alpha\beta} + \frac{1}{2} \operatorname{Ric}_{\alpha}^{\alpha} \wedge {}^*\operatorname{Ric}_{\beta}^{\beta} \\ & \equiv -L_{\text{SKY}} \\ & \quad - 2 \left(\operatorname{Ric}_{\alpha\beta} \wedge {}^*\operatorname{Ric}^{\alpha\beta} - \frac{1}{4} \operatorname{Ric}_{\alpha}^{\alpha} \wedge {}^*\operatorname{Ric}_{\beta}^{\beta} \right) \end{aligned}$	$\begin{aligned} & L_{\text{Euler}} := (-1)^{\text{sig}+1} \operatorname{Tr} \{ \Omega \wedge \Omega^{(*)} \} = \frac{(-1)^{\text{sig}}}{2} R^{\alpha\beta} \wedge R_{\alpha\beta}^{(*)} \\ & = dC_{\text{RR}^{(*)}} = \frac{(-1)^{\text{sig}}}{2} d \left(\Gamma_{\alpha\beta} \wedge R^{\alpha\beta(*)} - \frac{1}{3} \Gamma_{\alpha}^{\beta(*)} \wedge \Gamma_{\beta}^{\gamma} \wedge \Gamma_{\gamma}^{\alpha} \right) \\ & \equiv \frac{1}{2} R_{\alpha\beta} \wedge {}^*R^{\alpha\beta} - 2 \operatorname{Ric}_{\alpha\beta} \wedge {}^*\operatorname{Ric}^{\alpha\beta} + \frac{1}{2} \operatorname{Ric}_{\alpha}^{\alpha} \wedge {}^*\operatorname{Ric}_{\beta}^{\beta} \\ & \equiv -L_{\text{SKY}} \\ & \quad - 2 \left(\operatorname{Ric}_{\alpha\beta} \wedge {}^*\operatorname{Ric}^{\alpha\beta} - \frac{1}{4} \operatorname{Ric}_{\alpha}^{\alpha} \wedge {}^*\operatorname{Ric}_{\beta}^{\beta} \right) \end{aligned}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					